# Boundedness, Not the Posterior: A Preregistered Replication of TreeOPSD on Qwen3-4B

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**Abstract**

We preregister and run a replication of TreeOPSD (*Tree-Aware Posterior Distillation for Long-Horizon Agentic Reinforcement Learning*, arXiv:2610.23181) on Qwen3-4B-Thinking-2507 with search-agent training. Against the paper's own control, unclipped TreePO, TreeOPSD passes both of our preregistered bars at three seeds: pooled exact-match gaps of $+3.76$ $[+2.81, +4.74]$ on a four-set multi-hop QA average and $+3.45$ $[+2.13, +4.78]$ on the long-horizon BC+ benchmark (830 questions). Both passes are fragile: seed-level intervals at $n = 3$ include zero, and the largest contributor is one control seed that began to destabilise at the paper's dose. Against an added control, TreePO with ordinary ratio clipping, the BC+ gap is $+0.08$ $[-1.24, +1.41]$ and the QA gap is $+1.98$ $[+1.03, +2.94]$ pooled but declining across seeds $(3.78, 2.10, 0.05)$. The posterior term changes the per-update target by a median 1.05% at the paper's dose; a preregistered dose sweep that raises this drift sevenfold produces no growing accuracy gap, and at $4\times$ dose the unclipped control collapses in training in two of two seeds while both bounded variants train through. The budget match claimed for tree sampling holds in generated tokens ($1.03\times$ GRPO) but not in trained tokens ($1.48\times$). Best-checkpoint selection shrinks the gap toward zero. We conclude that the supported mechanism in this setting is update boundedness, not posterior credit assignment, and that bounded tree backup itself is an effective and, once clipped, low-variance (0.80 EM seed sd) method for training a 4B search agent.

## 1 Introduction

TreeOPSD [1] trains long-horizon agents by sampling a tree of continuations from shared prefixes, backing terminal rewards up the tree into turn-level values, converting those values into a bounded teacher distribution relative to a frozen behaviour policy, and matching the student to the teacher with a forward-KL self-distillation loss. The method is critic-free and is presented as a solution to credit assignment under sparse terminal rewards. Reported effects on multi-hop QA for Qwen3-4B-Thinking-2507 are small (TreeOPSD 44.13 vs. TreePO 43.40 vs. GRPO 43.37 MHQA average, five-seed means), with larger effects ($+2.9$ to $+4.0$) on a deep-research track.

A replication of this claim faces a confound in the original design. The paper's tree control, TreePO, applies its backed-up values without any ratio clip or bound, whereas TreeOPSD's teacher is bounded by construction. Outperformance against that control is therefore consistent with two mechanisms: (i) the posterior teacher supplies turn-level credit signal, as claimed; or (ii) any bound on the update improves stability and downstream accuracy. We resolve the confound by adding a control arm that keeps TreePO's sampler and backup but applies ordinary ratio clipping. We preregistered all arms, metrics, checkpoints and pass/kill bars before training (Section 2); the

preregistration (`PREREG-EXP1.md`, one pre-training amendment) ships with our code and result packages.

**Contributions.** (1) A five-arm, three-seed replication with frozen retrieval and a fixed evaluation checkpoint. (2) A direct measurement of the posterior term's per-update effect (M1) and a preregistered dose–response test of its causal role. (3) Evidence that unclipped tree backup is systematically unstable (collapse in 2/2 seeds at 4× dose; destabilisation in 1/3 seeds at 1×), and that either bound prevents it. (4) A token-accounting audit showing that the tree budget match holds for generated tokens (1.03×) but not for trained tokens (1.48×).

## 2 Design

**Model and task.** Qwen3-4B-Thinking-2507, ReAct-style search agents with a frozen retrieval backend, trained for 100 steps $(M, N) = (4, 2)$ trees for the tree arms) and evaluated greedily, one sample per question, at the fixed step-100 checkpoint. Tier 1 is the unweighted mean exact match (EM) over HotpotQA dev (7,405 questions), 2WikiMultihopQA, MuSiQue and Bamboogle (125). Tier 2 is BC+ (830 questions), our closed-corpus long-horizon track and the preregistered headline. Training and full evaluation ran on two DGX Station GB300 systems and an RTX PRO 6000 sidecar; all C-vs-E scores reported as headline were re-scored on GB300 hardware so the deciding comparison shares one device class (as-scored values differ by at most 0.49 EM and change no conclusion).

**Arms.** A: GRPO, $G = 8$ (seed 0). B: TreePO literal, the paper's control, likelihood-$\omega$ backup, prompt-normalised, no ratio or clip (seeds 0–2). C: TreeOPSD (seeds 0–2). D: GRPO, $G = 12$, the trained-token-matched GRPO (seed 0). E: TreePO with ordinary ratio clipping (seeds 0–2). All tree arms share the sampler, tree shape, data and retrieval.

**Statistics.** Rows are paired across runs by prompt hash. Gaps carry paired bootstrap intervals (10,000 resamples over questions, stratified by benchmark; pooled gaps average the per-question difference over seeds before resampling). We also report seed-level means, standard deviations and $t$ intervals ($n = 3$). The bootstrap intervals carry evaluation noise only; the seed intervals carry training-seed noise only. We report both throughout, because in this setting they disagree, which is itself a result.

**Preregistered bars.** F1 (posterior does the work): kill if $|\text{TreeOPSD} - \text{TreePO}| < 0.25$ with the interval excluding the paper's +0.73 on tier 1, or if the gap is $< 1.0$ on tier 2 (where the paper claims +2.90); pass at $\geq +0.5$ (tier 1) / $\geq +1.5$ (tier 2) with intervals excluding zero. Mechanism check M1: let drift_rel $= \overline{|\Gamma - V|}/\overline{|V|}$ over inner-loop blocks 2–4, where $\Gamma$ and $V$ are the TreeOPSD and TreePO targets. If drift_rel $< 0.02$, the posterior term is declared inactive at that dose whatever the EM shows, and a preregistered dose sweep (EXP-1c) follows. F2 (budget): kill if any of generated tokens, tool calls or GPU-hours exceeds GRPO by $> 1.15\times$, or if budget matching removes $> 50\%$ of the gap. F3 (selection): kill if the gap's interval includes zero at the fixed checkpoint but excludes zero under best-in-window selection. F4 (retrieval): frozen by construction. F5 (horizon): gain should grow with hop count; our MHQA design caps at five turns, so F5 evidence here is weak by construction and we claim no trend.

Table 1: Step-100 exact match. Tier 1 (QA avg) is the mean over the four QA sets; Bamboogle (125 questions) is included in the average but its column is omitted for width. C is shown GB300-scored (see text). A and D ran seed 0 only. The untrained base scores 30.8 on frozen HotpotQA dev (paper ReAct 31.61) and 22.8/6.5 on the 500/200-question interim subsets.

| Run | BC+ | HotpotQA | 2Wiki | MuSiQue | |
|---|---|---|---|---|---|
| A GRPO G=8 s0 | 10.00 | 43.44 | 48.55 | 18.41 | |
| B TreePO s0/s1/s2 | 10.48/7.23/6.27 | 42.58/45.97/39.54 | 49.93/53.78/41.57 | 18.57/20.61/14.05 | 39 |
| C TreeOPSD s0/s1/s2 | 10.48/9.04/13.61 | 46.71/46.29/45.43 | 51.67/50.65/48.74 | 22.35/21.10/21.27 | 43 |
| D GRPO G=12 s0 | 7.83 | 44.09 | 51.19 | 20.32 | |
| E TreePO-clipped s0/s1/s2 | 12.29/12.05/8.55 | 44.74/45.12/44.11 | 47.73/48.24/49.48 | 20.36/19.49/18.45 | 39 |

Table 2: Pooled gaps (question bootstrap, 10,000 resamples) with per-seed point values and seed-level $t$ intervals ($n = 3$). C−E is hardware-matched; as-scored it reads $+2.17\,[+1.29, +3.06]$ (tier 1) and $+0.48\,[-0.84, +1.81]$ (tier 2).

| Gap | Tier | Pooled [95% CI] | Per-seed | Seed $t$ interval |
|---|---|---|---|---|
| C−E | QA | $+1.98$ $[+1.03, +2.94]$ | $3.78, 2.10, 0.05$ | $[-2.66, +6.61]$ |
| C−E | BC+ | $+0.08$ $[-1.24, +1.41]$ | $-1.81, -3.01, 5.06$ | $[-10.74, +10.90]$ |
| C−B | QA | $+3.76$ $[+2.81, +4.74]$ | $4.50, 0.31, 6.47$ | $[-4.06, +11.57]$ |
| C−B | BC+ | $+3.45$ $[+2.13, +4.78]$ | $-0.36, 3.37, 7.35$ | $[-6.13, +13.03]$ |
| C−A (s0) | QA / BC+ | $+3.78$ $[+2.41, +5.21]$ / $+0.48$ $[-1.69, +2.65]$ | – | – |
| C−D (s0) | QA / BC+ | $+2.08$ $[+0.69, +3.53]$ / $+2.65$ $[+0.60, +4.82]$ | – | – |

## 3 Results

**Posterior vs. bounded (C−E): the deciding comparison.** Table 2 gives the hardware-matched result. On tier 1 QA the pooled gap is $+1.98$ $[+1.03, +2.94]$, meeting the script's pass bar; per-seed it declines $3.78, 2.10, 0.05$, and the seed-level interval $[-2.66, +6.61]$ includes zero. On tier 2 BC+, where the paper's large effect is claimed, the pooled gap is $+0.08$ $[-1.24, +1.41]$ (per-seed $-1.81, -3.01, +5.06$), meeting the kill bar. The clipped control's tier 1 seed sd (0.80) is the smallest of any arm (C 1.07, B 3.95). We read tier 1 as a real but seed-fragile pooled edge of at most $\approx$2 EM whose cause Section 4 does not allow us to credit to the posterior mechanism, and tier 2 as no posterior-specific effect.

**Against the paper's control (C−B).** Pooled gaps are tier 1 $+3.76$ $[+2.81, +4.74]$ and tier 2 $+3.45$ $[+2.13, +4.78]$: passes by the letter of F1's EM bars. Seed-level intervals ($[-4.06, +11.57]$ and $[-6.13, +13.03]$) include zero. B seed 2 began to destabilise in its final 10 steps at the paper's dose: reward fell from 0.54–0.70 to 0.22–0.40, the length-clip ratio rose from 0.00 to 0.27, and the checkpoint issued 3.52 tool calls per BC+ question against 1.81/1.97 for B s0/s1. Excluding B s2, the two-seed gaps are $+2.40$ (tier 1) and $+1.51$ (tier 2). Clipping alone (E−B) improves BC+ in all three seeds ($+1.81, +4.82, +2.29$).

## 4 Mechanism: M1 and the dose response

At the paper's dose ($E = 4$ inner-loop blocks per step) M1 trips: drift_rel median 1.05% over the full run and 1.56% in the last quarter, below the preregistered 2% inactivity line. Backup weights are near-one-hot in both tree arms ($\omega_{\text{peaked}} \approx 0.85$–$0.91$), so the backup is effectively single-child in this setting. EXP-1c raised the dose. At $E = 8$, drift rose (median 1.23%, last-quarter 1.89%) and the tier 1 subset gap was $-0.8$ EM: no growing accuracy gap. At $E = 16$ (drift 4.79%/7.74%, 34% of calls above 10%) unclipped TreePO collapsed in training: reward 0.002 and length-clip ratio 0.99 by step 33; the step-40 checkpoint scores 0.0/0.0 on the interim subsets. A second seed collapsed the same way (subset scores 17.8/0.0; BC+ fully collapsed, QA degraded but answering). At the same dose, clipped TreePO trained through (reward 0.49–0.64, clip ratio $\leq 0.10$; subsets 34.2/8.5) as did TreeOPSD (0.50–0.69; 35.4/5.0). Boundedness, in either form, is what separates survival from collapse; the posterior term adds no measurable signal beyond it at any tested dose. Combined with B s2's destabilisation at $E = 4$, unclipped tree backup failed fully in 2/2 seeds at $4\times$ dose and partially in 1/3 seeds at $1\times$; neither bounded variant failed in any run.

**Budget (F2).** Tree sampling is $1.03\times$ GRPO $G$=8 in generated tokens per prompt and $1.48\times$ in trained tokens per prompt (41,064 vs. 27,779 in the matched smoke runs), because shared prefixes are re-scored per branch during training. Token-matched GRPO (arm D) removes 44% of the tier 1 C−A gap, below the 50% F2b kill line; on BC+ the $G$=8 gap is $\approx 0$ and the ratio is undefined. The claimed budget match therefore depends on accounting in generated tokens only.

**Selection (F3).** Best-in-window scoring over checkpoints 10/25/50/75/100 on fixed subsets shrinks C−B toward zero (per-seed mean +0.3 tier 1 / +0.5 tier 2, vs. $+1.5/+2.5$ at the final checkpoint on the same subsets), chiefly by rescuing B s2's pre-destabilisation peak (36.6 at step 75 vs. 32.8 at step 100, qa500i). The F3 kill pattern is not met and the shift runs in the opposite direction; selection favours the control. Our grid is coarser than the preregistration's every-5-steps ideal, which is conservative against this conclusion. Retrieval was frozen by construction (F4); an eval-drift control (B s0 Bamboogle re-scored after the study) returned an identical 45.6.

## 5 Discussion and limitations

Three claims survive this replication. Bounded tree backup trains a 4B search agent effectively (base 22.8 EM on our 500-question QA subset; $40.61 \pm 0.80$ and $42.58 \pm 1.07$ full-set QA means for arms E and C). Unclipped tree backup is fragile, systematically so at high dose. TreeOPSD outperforms the control as the paper defines it. The claim that does not survive is the mechanism: the posterior teacher's per-update effect is $\approx 1\%$, its dose response is flat, and against a bounded control its advantage vanishes on the long-horizon track. The residual tier 1 C−E pooled edge is consistent with second-order dynamics we observed but did not isolate (C sustained $\approx$43K trained tokens per prompt where B's fell to $\approx$33K, and learned a longer, search-heavier style: 14.2 vs. 10.6 tool calls per prompt late in training) and with seed noise; $n = 3$ cannot adjudicate. We also note the paper's reported seed sd ($\approx 0.17$ EM) is far smaller than any arm here (0.80–3.95); replications at single seeds in this setting should be read accordingly.

Limitations. Three seeds: every seed-level interval includes zero. One model at one scale. BC+ carries $\approx \pm 2$ EM per-arm eval noise at 830 questions; best-in-window used small subsets; GRPO arms ran one seed; eval is single-sample greedy; our tier 1 average (four sets) is composed differently from the paper's MHQA average, so levels are not directly comparable (our base-model harness

anchor matches theirs: 30.8 vs. 31.61). Nothing here speaks to TreeOPSD at larger scale, with a live retriever, or on horizons beyond our design.

## 6 Conclusion

Replicated against its own control, TreeOPSD passes. Replicated against a control that is merely clipped, its long-horizon advantage is zero and its QA advantage is at most about two points and seed-fragile, while its mechanism measures at $\approx 1\%$ of the update it is supposed to drive. The robust, reusable finding of this study is about the control: tree backup without a bound collapses, and any bound fixes it. Papers that compare a bounded method against an unbounded control should expect this confound to be checked.

**Reproducibility.** Preregistration, training and scoring code, per-run logs and the three seed-package JSONs underlying Tables 1–2 are released with this note at `https://github.com/sjronan/treeopsd-replication` (`PREREG-EXP1.md`, `scripts/seed_package.py`, `results/`). Experiments were designed and directed by the author and executed with AI-agent assistance under a file-based controller protocol; all rulings and deviations are preserved in the released correspondence log.

## References

[1] Zou, H., Wang, Q., et al. Tree-Aware Posterior Distillation for Long-Horizon Agentic Reinforcement Learning. arXiv:2610.23181, 2026.